

SUMMER QUEST V2

SAFE ARCHITECTURE EVOLUTION PLAN

Status: Production System

Important: Children are actively using the application.

Progress data already exists.

XP, Coins, Badges, Streaks, Attempts, Reading Logs, Chores, Parent Reviews already exist.

This plan must NOT break production.

GLOBAL SAFETY RULE

DO NOT MODIFY

prisma/schema.prisma

database structure

existing routes

existing APIs

existing lesson logic

existing quiz logic

existing progress logic

existing reward logic

existing approval logic

unless explicitly approved in a future migration phase.

This program is currently:

Documentation Refactor Architecture Refactor Knowledge Refactor

ONLY.

PHASE 0 PRODUCTION SNAPSHOT

Goal:

Protect existing system.

Tasks:

1. Backup:

prisma/dev.db

2. Export:

Current Docs Inventory

3. Export:

Current Database Models

4. Create:

ARCHITECTURE_BASELINE.md

Content:

Current state

Current schema

Current routes

Current modules

Current limitations

Acceptance:

No code changes.

No database changes.

No runtime changes.

Risk:

0%

PHASE 1 KNOWLEDGE ROUTER SYSTEM

Goal:

Reduce AI onboarding tokens.

Current:

Claude reads many documents.

Future:

Claude reads one router.

Tasks:

Create:

docs/

00_AGENT_INDEX.md

01_RULES.md

KNOWLEDGE_MAP.json

00_AGENT_INDEX.md

Must be:

<5KB

Purpose:

Single entry point.

Document router.

Task router.

Knowledge router.

AI onboarding router.

01_RULES.md

Must contain:

Most important rules only.

Compressed version.

No long explanations.

KNOWLEDGE_MAP.json

Maps:

UI

Database

Content

Modules

Bugs

Architecture

to documents.

Acceptance:

AI can identify required docs without reading all docs.

Expected token reduction:

50-80%

Risk:

0%

PHASE 2 DOCUMENT RESTRUCTURE

Goal:

Create future architecture docs.

Tasks:

Create:

AI_DATA_STANDARDS.md

MODULE_ARCHITECTURE.md

CURRICULUM_STRUCTURE.md

CONTENT_IMPORT.md

ASSET_MODEL.md

RAG_ARCHITECTURE.md

Update:

AGENTS.md

with new onboarding flow.

Acceptance:

New docs exist.

Old docs remain unchanged.

No runtime changes.

Risk:

0%

PHASE 3 MODULE ARCHITECTURE DESIGN

Goal:

Prepare subject isolation.

DO NOT TOUCH CODE.

Create:

modules/

english/

math/

vietnamese/

science/

Inside each module:

MODULE.md

README.md

content/

assets/

games/

importers/

Example:

modules/english/MODULE.md

Contains:

Scope

Ownership

Content Types

Future Games

Import Rules

Acceptance:

Module folders exist.

Documentation exists.

No runtime changes.

Risk:

0%

PHASE 4 CURRICULUM STRUCTURE

Goal:

Support Grades 3-6+

Tasks:

Create:

Curriculum hierarchy.

Example:

Grade 3

Math

Vietnamese

English

Science

Grade 4

Math

Vietnamese

English

Science

Grade 5

Math

Vietnamese

English

Science

Acceptance:

Curriculum documented.

No code changes.

Risk:

0%

PHASE 5 PDF IMPORT FOUNDATION

Goal:

Prepare for Smart Start 4.

Prepare for Math Grade 5 books.

Prepare for future books.

Tasks:

Create:

CONTENT_IMPORT.md

Rules:

PDF processed once.

Never re-read PDFs.

Pipeline:

PDF

↓

OCR

↓

Image Extraction

↓

Knowledge Extraction

↓

JSON

↓

Database

↓

Vector Store

Acceptance:

Import standard documented.

No code changes.

Risk:

0%

PHASE 6 CONTENT REPOSITORY

Goal:

Move future content outside runtime code.

Create:

content_repository/

english/

math/

vietnamese/

science/

Store:

JSON only.

No PDFs.

No DOCX.

No raw content.

Acceptance:

Repository structure exists.

No runtime changes.

Risk:

0%

PHASE 7 ASSET REPOSITORY

Goal:

Prepare reusable educational assets.

Create:

assets/

english/

math/

vietnamese/

science/

Rules:

Every asset has ID.

Every lesson references IDs.

Never embed images directly.

Acceptance:

Asset standard exists.

No runtime changes.

Risk:

0%

PHASE 8 RAG FOUNDATION

Goal:

Future AI retrieval.

Tasks:

Create:

RAG_ARCHITECTURE.md

Rules:

Chunk Size

200-500 words

Retrieval:

Lesson Chunk

Question Chunk

Vocabulary Chunk

Objective Chunk

Never:

Retrieve entire books.

Acceptance:

Architecture documented.

No runtime changes.

Risk:

0%

PHASE 9 CONTENT MIGRATION PREPARATION

Goal:

Prepare future migration.

DO NOT MIGRATE.

Tasks:

Create:

Migration Plan Only.

Map:

Current Lesson

↓

Future Book

Unit

Lesson

Structure

Acceptance:

Migration strategy exists.

No code changes.

No database changes.

Risk:

0%

PHASE 10 OPTIONAL FUTURE NOT NOW

ONLY AFTER SYSTEM STABILIZATION

Possible future work:

Database redesign

Book table

Unit table

Objective table

Vocabulary table

Asset table

Vector DB

Module Runtime Loading

Plugin Architecture

Content API

This phase is NOT approved.

This phase requires:

Separate review.

Separate migration plan.

Separate database backup.

Separate testing cycle.

FINAL SUCCESS CRITERIA

1. Production remains functional.
2. Children keep learning normally.
3. No data loss.
4. No schema changes.
5. No runtime behavior changes.
6. AI onboarding token cost reduced.

7. Future module architecture ready.
8. Future PDF import architecture ready.
9. Future RAG architecture ready.
10. Future curriculum expansion ready.